
EMCH 574 – Vibration and Waves

Homework 01

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Due date TBD

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items
- UG students are not required to do the G work, but they may do it for bonus points
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#)
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga...) [reference](#)
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit
- Normalized plots are plots with the [y-axis](#) scaled between -1 and +1. A plotted variable is normalized by dividing it by its maximum absolute value

Notations and Abbreviations

- BVP = boundary value problem
- EOM = equation of motion
- FBD = free body diagram
- FRF = frequency response function
- IVP = initial value problem
- LHS = left hand side
- NLM = Newton's law of motion
- ODE = ordinary differential equation
- PDE = partial differential equation
- RHS = right hand side

1 Pendulum

A.1 Analyze Pendulum

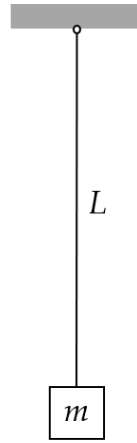


Figure 1: pendulum of length L and mass m

Consider a pendulum of length L and mass m as shown in Figure 1. Do the following:

- (1) Draw the pendulum displaced by an angle θ from the vertical
- (2) Draw free-body diagram (FBD)
- (3) Write Newton's law of motion (NLM) equation
- (4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (5) Deduce the governing equation.
- (6) Assume $\theta(t) = e^{st}$ and substitute into the governing equation
- (7) Deduce and solve the characteristic equation
- (8) How are the roots $s_{1,2}$? Choose one of the following options:
 - (a) real
 - (b) imaginary
 - (c) complex
- (9) Define the natural frequency ω_n
- (10) Express the solutions s_1 and s_2 of the characteristic equation in terms of ω_n
- (11) Write the solution in terms of exponential functions of $\omega_n t$
- (12) Write the solution in terms of trigonometric functions of $\omega_n t$
- (13) write the Hz frequency f_n
- (14) write the period of oscillation τ
- (15) write the particle motion $u(t)$ in terms of

- (a) exponential functions
- (b) trigonometric functions

(16) solve the initial value problem (IVP) and find the response $u(t)$ for:

- (i) initial displacement u_0
- (ii) initial velocity $\dot{u}_0$
- (iii) simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$

A.2 Calculate Pendulum Response To Initial Displacement

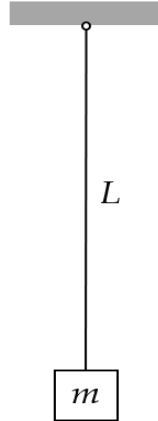


Figure 2: pendulum of length L and mass m

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 565 \text{ mm}$ and mass $m = 52 \text{ kg}$ as shown in Figure 2. When manufactured in series production, the pendulum length values are not exact due to manufacturing tolerance. The mm length values for ten pendulum sets are: $L = [561.535, 562.691, 563.845, 563.845, 565.0, 565.0, 566.155, 566.155, 567.309, 568.464] \text{ mm}$. Do the following:

- (1) For the ten pendulum sets, calculate
 - (a) rad/sec natural frequency ω_n
 - (b) Hz natural frequency f_n
 - (c) period of oscillation τ
 - (d) display in table format L, ω_n, f_n, τ
- (2) Mean values: calculate and display the mean values of ω_n, f_n, τ
- (3) Initial value problem (IVP): assume initial displacement $u_0 = 2 \text{ mm}$ and zero initial velocity. Use the mean values of ω_n, f_n, τ to do the following:
 - (a) Calculate and display the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$
 - (b) Calculate and display the duration t_{Nosc} of N_{osc} oscillations and the corresponding position of the mass u_{Nosc} for $N_{\text{osc}} = 10$
 - (c) Plot the response of the system for N_{osc} oscillations and use a datatip on the plot to find the value $u(t_1)$ at t_1

A.3 Calculate Pendulum Response To Impact

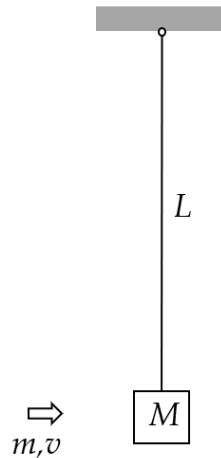


Figure 3: impact excitation of a pendulum system

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 565 \text{ mm}$ and mass $M = 52 \text{ kg}$ as shown in Figure 3. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 22 \text{ g}$ and velocity $v = 945 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate. Do the following:

- (1) display input data
- (2) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it
- (3) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (4) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$
- (5) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) the largest response
 - (b) the response value calculated at item (4)
- (6) Discuss your results

2 Spring-mass-damper vibrating systems

B.1 Analyze Spring-Mass Vibrating System



Figure 4: spring-mass vibrating system

Consider a spring-mass vibrating system as shown in Figure 4. Do the following:

- (1) Draw free-body diagram (FBD)
- (2) Write NLM equation
- (3) Deduce the equation of motion (EOM)
- (4) Deduce the governing equation. Clearly state your assumptions.
- (5) Assume $u(t) = e^{st}$ and substitute into the governing equation
- (6) Deduce the characteristic equation
- (7) Define the natural frequency ω_n in terms of m, k
- (8) Solve the characteristic equation and find $s_{1,2}$ in terms of ω_n
- (9) Answer the following question: How are the roots $s_{1,2}$? Choose one of the following options:
 - (a) real
 - (b) imaginary
 - (c) complex
- (10) Write the solution in terms of exponential functions
- (11) Write the solution in terms of trigonometric functions
- (12) Solve the initial value problem (IVP) and find the response $u(t)$ for:
 - (i) initial displacement u_0
 - (ii) initial velocity $\dot{u}_0$
 - (iii) simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$

B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring, k , mass, m , and viscous damper, c , as presented in Figure 5. Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

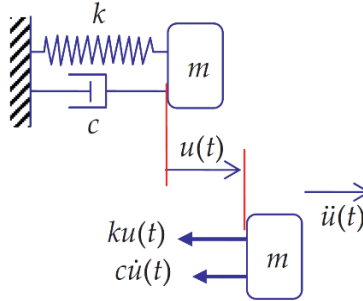


Figure 5 mass-spring-damper vibrating system

Do the following:

- (1) State your assumptions in performing this analysis
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k
- (5) Assume $u(t) = e^{st}$ and substitute into EOM
- (6) Deduce the characteristic equation
- (7) Find the roots $s_{1,2}$ of the characteristic equation
- (8) Define critical damping c_{cr} in terms of m and k
- (9) Define damping ratio ζ
- (10) Discuss the values of ζ for the three damping cases:
 - (a) underdamped
 - (b) critically damped
 - (c) overdamped
- (11) Express the physical damping c in terms of damping ratio ζ , mass m , stiffness k
- (12) Define the natural frequency ω_n in terms of m, k
- (13) Express in terms of ω_n , ζ the $s_{1,2}$ roots found at (7)
- (14) Assume underdamped system and express the $s_{1,2}$ roots of (13) as complex numbers in terms of ω_n and ζ
- (15) Define the damped frequency ω_d
- (16) Substitute (15) into (14) and express the roots s_1 and s_2 in terms of ω_n , ω_d , ζ
- (17) Assume underdamped conditions and state the inequality that ζ should satisfy

- (18) Write the general solution of the ODE deduced at (4) in terms of some constants C_1, C_2 multiplying exponential functions of s_1, s_2
- (19) Use (16) to write the solution (18) in terms of exponential functions of $\omega_n, \omega_d, \zeta$
- (20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials
- (21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants C and ϕ to be determined from the initial conditions.
- (22) Sketch the functions discussed at (21)
- (23) Solve the initial value problem (IVP) for $\zeta < 1$ and find the response $u(t)$ for initial displacement u_0 with zero initial velocity
- (24) Solve the initial value problem (IVP) and find the response $u(t)$ for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$

B.3 Calculate Spring-Mass-Damper Response To Initial Displacement

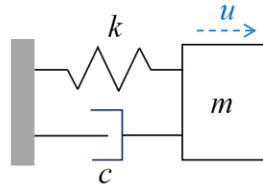


Figure 6: spring-mass-damper vibrating system

Consider a spring-mass vibrating system with $m = 12 \text{ kg}$, $k = 655 \text{ N/m}$, $c = 19 \text{ N/(m/s)}$ as shown in Figure 6. The vibrating system is given an initial displacement $u_0 = 3.5 \text{ mm}$ with zero initial velocity. Do the following:

- (1) Display input data
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n
- (3) Calculate the critical damping c_{cr}
- (4) Calculate the damping ratio ζ
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d
- (6) Calculate the damped period of oscillation τ_d
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2
- (8) Calculate the constants A , B and then the constants C , ϕ
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement
- (11) Plot the response of the system for $N_{osc} = 10$ oscillations and use datatips on the plot to find the following:
 - (a) the largest response
 - (b) the response value calculated at item (9)
 - (c) the time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10)
- (12) Discuss your results

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 7.

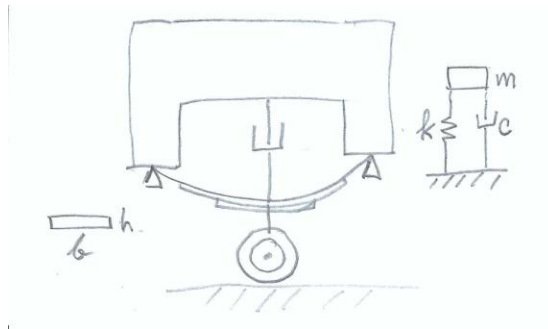


Figure 7 Truck suspension consisting of a leaf spring and an oil damper

The leaf spring can be approximated by a uniform beam of length $L = 830 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$, $b = 65 \text{ mm}$ made of steel with $E = 200 \text{ GPa}$. The truck mass apporportioned to the wheel can be estimated as $m = 457 \text{ kg}$. Assume initial displacement $u_0 = 125 \text{ mm}$ and zero initial velocity $\dot{u}_0 = 0$. Do the following:

- (1) Display input data
- (2) Calculate the second moment of area I of the beam cross- section
- (3) Calculate the flexural stiffness EI of the beam
- (4) Calculate the spring constant k of the system
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n
- (6) Calculate the critical damping c_{cr}
- (7) Assume 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c . Print ζ and c
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d
- (9) Calculate the damped period of oscillation τ_d
- (10) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2
- (11) Calculate the constants A , B and then the constants C , ϕ
- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement
- (14) Plot the response of the system for $N_{osc} = 10$ oscillations and use datatips on the plot to find the following:
 - (a) the largest response
 - (b) the response value calculated at item (11)
 - (a) the time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13)

- (15) Discuss your results
- (16) Graduate work: assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.
- (a) Repeat the calculation of items (6) through (12)
 - (b) Compare and discuss the two results

B.5 Overdamped Spring-Mass-Damper Systems

Consider a spring-mass-damper system with natural frequency $f_n = 6.25 \text{ Hz}$ and various values of the damping ratio ζ . The system is given an initial displacement $u_0 = 13 \text{ mm}$ with zero initial velocity. Do the following:

- (1) display input data
- (2) calculate the rad/s frequency ω_n
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) write the expressions of the roots s_1, s_2 in terms of ω_n and ζ
 - (b) write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2
 - (c) write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$
 - (d) take $\zeta = 2.3$ and calculate the numerical values of s_1, s_2 and

C_1, C_2

- (e) plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$
- (4) Assume critically damped conditions $\zeta = 1$ and do the following:
 - (a) write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n
 - (b) write the expression of the complete solution in terms of ω_n and C_1, C_2
 - (c) write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$
 - (d) calculate the numerical values of C_1, C_2
 - (e) plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$
- (5) Graduate Work: derive the ODE for $\zeta = 1$ and verify that the two solutions $u_1(t)$ and $u_2(t)$ from (4)(a) satisfy the ODE
- (6) Graduate Work: some race-car manufacturers design for critical damping whereas others use overdamped design. Discuss the pros and cons of the two options

B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time- dependent force $F(t)$ as shown in Figure 8.

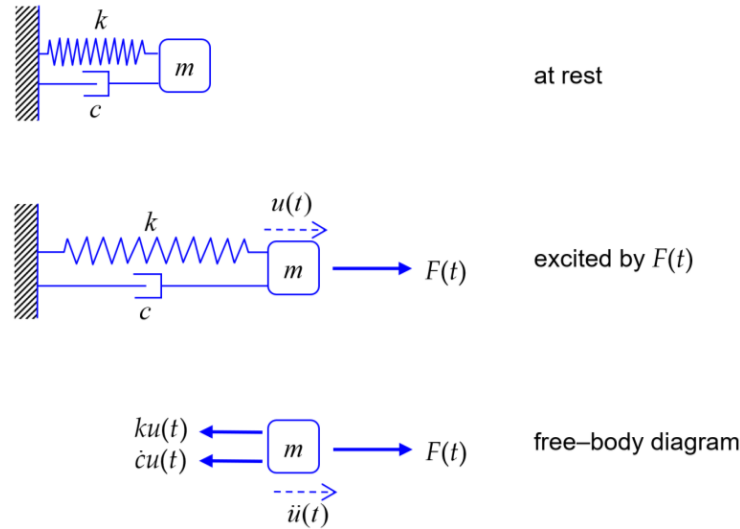


Figure 8 spring-mass-damper system excited by force $F(t)$

Do the following:

- (1) State your assumptions in performing this analysis
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k , $F(t)$
- (5) Assume harmonic excitation $F(t) = \hat{F} e^{i\omega t}$, substitute in (4), and divide by m
- (6) Define the normalized excitation amplitude $\hat{f} = \hat{F}/m$ and recall the definitions of ω_n and ζ to obtain a nonhomogeneous linear ODE in ω_n , ζ , $\hat{f}$
- (7) Write the general solution $u(t)$ as a sum of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$
- (8) Write the general form of the complementary solution $u_c(t)$
- (9) Find the particular solution $u_p(t)$
- (10) Use (8), (9) to write the complete solution $u(t)$; discuss the meaning of transient response and steady-state response.
- (11) Find an expression for the steady-state response amplitude $\hat{u}_{ss}$ in terms of ω_n , ζ , $\hat{f}$, ω
- (12) Develop the expression for the frequency response function $\text{FRF}(\omega)$ in terms of ω_n , ζ , ω
- (13) Discuss the variation with frequency of FRF magnitude and phase
- (14) Give the definition of the magnification factor $M(\omega)$

- (15) Give the expression for the magnification factor $M(\omega)$ in terms of ω_n and ζ
- (16) Give the definition of the resonance frequency ω_r
- (17) Give the expression for the resonance frequency ω_r in terms of ω_n and ζ
- (18) Give the expression of the peak magnification M_r
- (19) Give the expression of the FRF phase $\phi(p)$ where $p = \omega/\omega_n$
- (20) Give the definition of phase resonance and deduce the FRF expression at phase resonance
- (21) Calculate the magnification factor at phase resonance
- (22) Calculate the magnitude of the physical response at phase resonance
- (23) Discuss the difference between true resonance and phase resonance
- (24) Write and compare the expressions for the resonance frequency ω_r and damped frequency ω_d
- (25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency ω

B.7 Calculate Flex-Beam Response To Slightly Unbalanced Electric Motor Excitation

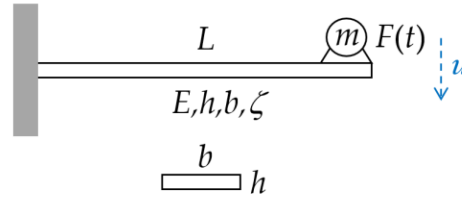


Figure 9 damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor

Consider a cantilever beam of length $L = 568 \text{ mm}$, elastic modulus $E = 200 \text{ GPa}$, cross-section properties $h = 3 \text{ mm}$, $b = 37 \text{ mm}$ and damping ratio $\zeta = 14\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.52 \text{ kg}$ operating at $1,500 \text{ RPM}$ is attached at the beam tip as shown in Figure 9. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F} e^{i\omega t}$ with $\hat{F} = 12 \text{ mN}$. The excitation frequency ω depends on the electric motor RPM. Do the following:

- (1) Display input data
- (2) Calculate
 - (a) cross-section second moment of area I , flexural stiffness EI , and spring constant k
 - (b) natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c
 - (c) quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$
- (3) Calculate the values of
 - (a) resonant frequency ω_r and damped frequency ω_d
 - (b) peak magnification M_r
 - (c) magnification at phase resonance M_{90}
 - (d) physical response $\hat{u}_{90}$ at phase resonance and its magnitude $\hat{u}_{90}$
 - (e) calculate the Hz frequencies f_r , f_n and the corresponding RPM values
- (4) assume excitation at phase resonance:
 - (a) calculate the excitation frequency ω in rad/s and f in Hz
 - (b) calculate period of oscillation τ
 - (c) Plot the time response for $N_{osc} = 10$ oscillations. Hint: plot the real part of the complex displacement $u(t)$
- (5) Graduate work:
 - (a) calculate and display the operational frequency f_0 in Hz and the corresponding ω_0 in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to f_0 with $N_f = 5000$ points
 - (b) calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot

- (c) calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM
- (d) comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM

B.8 Graduate Work: Calculate Horizontal Flex-Beam Response To Impact

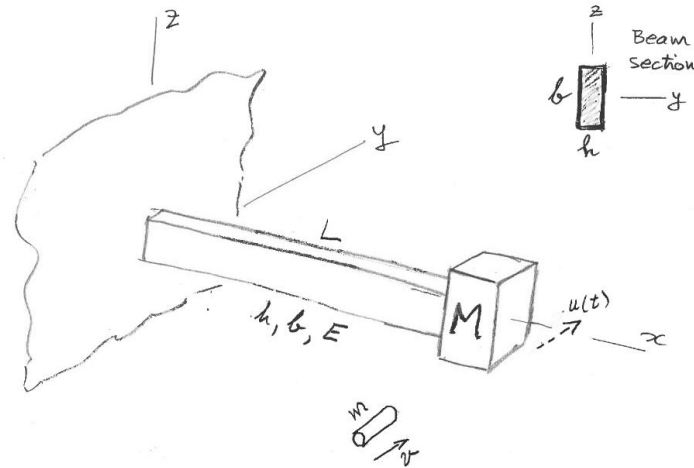


Figure 10: impact excitation of a horizontal flex-beam vibration system

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $M = 53 \text{ kg}$ attached at its tip as shown in Figure 10. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The beam is initially at rest. A small projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane. Do the following:

- (1) Display input data
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$
- (3) Calculate the cross-sectional stiffness EI of the beam
- (4) Calculate the flex-beam spring constant k
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ
- (7) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) the largest response
 - (b) the response calculated at item (7)
- (9) Discuss your results

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Vertical Spring-Mass Vibration System

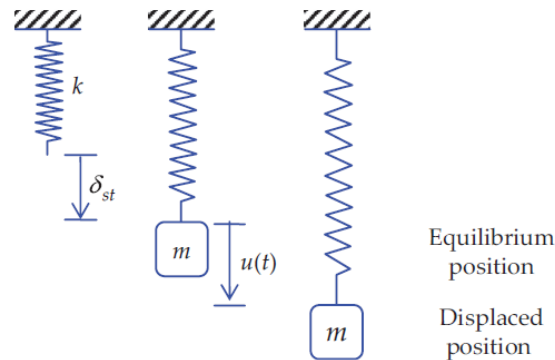


Figure 11 vertical spring-mass vibration system

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 11. Do the following:

- (1) Assume static equilibrium
 - (a) Calculate the weight W acting on the mass m
 - (b) Draw free-body diagram (FBD)
 - (c) Write equilibrium equation
 - (d) Solve for the static displacement δ_{st}
- (2) Assume vibration with time-varying vertical displacement $u(t)$
 - (a) Draw free-body diagram (FBD)
 - (b) Deduce equation of motion (EOM)
 - (c) Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1
- (3) Graduate Work: comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ wrt vertical axis.

C.2 Analyze Vertical Flex-Beam Vibration

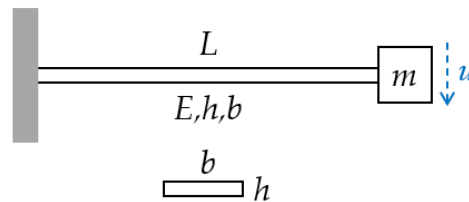


Figure 12: vertical flex-beam vibrating system

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 12. The beam cross-section properties are E , h , b . Do the following:

- (1) Calculate the cross-sectional stiffness EI of the beam for vertical flexural deformation
- (2) Calculate the spring constant k of the system
- (3) Assume static equilibrium:
 - (a) Calculate the weight W acting on the mass m
 - (b) Draw free-body diagram (FBD)
 - (c) Write equilibrium equation
 - (d) Solve for the static displacement δ_{st}
- (4) Assume vibration with time-varying vertical displacement $u(t)$ measured from the static equilibrium position:
 - (a) Draw free-body diagram (FBD)
 - (b) Deduce equation of motion (EOM)
 - (c) Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4
- (5) Graduate Work: comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ wrt vertical axis.

C.3 Analyze Horizontal Flex-Beam Vibration

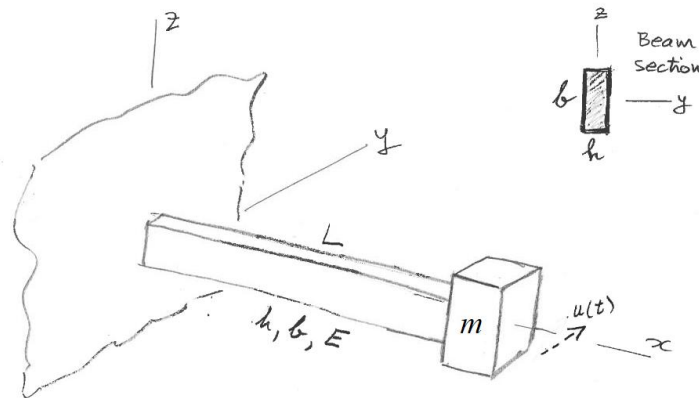


Figure 13: horizontal flex-beam vibrating system

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 13. The beam cross-section properties are E , h , b . Do the following:

- (1) Assume vibration with horizontal time-varying displacement $u(t)$ as shown in Figure 13
- (2) Calculate the cross-sectional stiffness EI of the beam for flexural deformation consistent with $u(t)$
- (3) Calculate the flex-beam spring constant k
- (4) Draw free-body diagram (FBD) including all the forces acting on the mass m
- (5) Deduce equation of motion (EOM)
- (6) Compare the EOM deduced here with the EOM derived in Problem B.1
- (7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$

C.4 Calculate Horizontal Flex-Beam Response To Initial Displacement

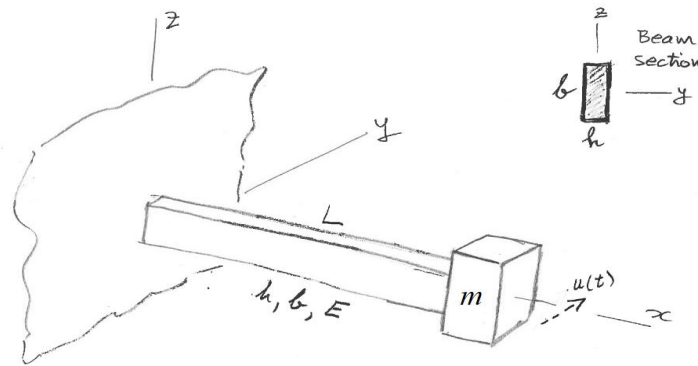


Figure 14: flex-beam vibrating system

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $m = 5.35 \text{ kg}$ attached at its tip as shown in Figure 14. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The tip mass is given an initial displacement $u_0 = 5.4 \text{ mm}$ with zero initial velocity. Do the following:

- (1) Display input data
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$
- (3) Calculate the cross-sectional stiffness EI of the beam
- (4) Calculate the spring constant k of the system
- (5) Calculate the natural frequency in rad/sec ω_n
- (6) Calculate the natural frequency in Hz f_n
- (7) Calculate the period of oscillation in sec τ
- (8) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$
- (9) Plot the response of the system for $N_{\text{osc}} = 25$ oscillations and use a data tip on the plot to find the value $u(t_1)$ at t_1 on the plot
- (10) Discuss your results

C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time- dependent force $F(t)$ as shown in Figure 15.

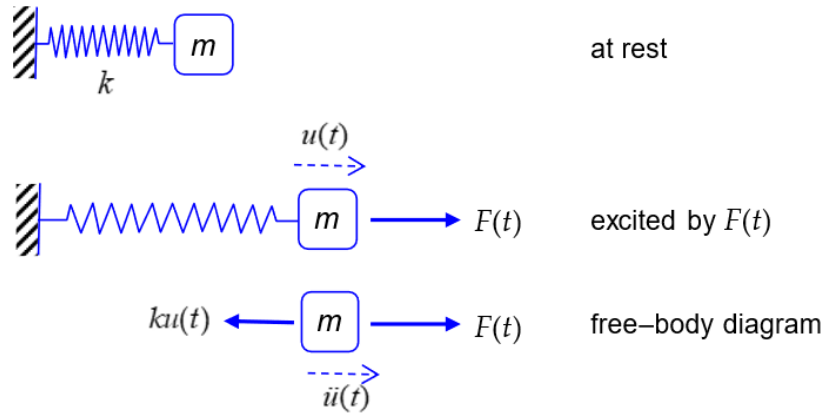


Figure 15 spring-mass system excited by force $F(t)$

General analysis of spring-mass forced vibration Do the following:

- (1) State your assumptions in performing this analysis
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation
- (4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of m , c , k , $F(t)$
- (5) Write the ODE general solution as the superposition of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$
- (6) Recall the complementary solution $u_c(t)$ as the solution of the homogeneous equation studied in HW01 and write it in terms of trigonometric functions multiplied by constants A and B .

Analysis of spring-mass response to suddenly applied load

- (7) Assume the loading $F(t)$ is a suddenly applied load of magnitude F_0
- (8) Find the particular solution for this case of a suddenly applied load
- (9) Write the complete solution as the sum of $u_c(t)$ from (6) and $u_p(t)$ from (8)
- (10) Assume “at rest” initial conditions to find the constants A and B and write the complete solution $u(t)$ as a function of F_0 , k , and $\cos \omega_n t$
- (11) Use (10) to find the largest displacement $u_{\max}$ and compare it with the static displacement u_{st} .

C.6 Calculate Cantilever Response To Suddenly Applied Weight

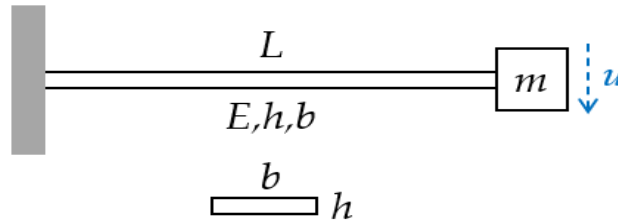


Figure 16: cantilever beam with tip mass

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 568 \text{ mm}$ and a mass $m = 5.35 \text{ kg}$ that has to be attached to its tip as shown in Figure 16. The beam is made of steel with elastic modulus $E = 200 \text{ GPa}$ and yield stress $Y = 710 \text{ MPa}$ cross-section properties $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. Do the following:

- (1) Display input data
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system
- (3) Calculate the weight F_0 of the mass m
- (4) Static case: assume the mass is attached onto the undeformed beam and then let go gradually without inducing any vibration and calculate the static displacement u_{st}
- (5) Dynamic case: assume the mass is attached onto the undeformed beam and then let go suddenly inducing a state of vibration:
 - (a) calculate the maximum dynamic displacement u_{dyn}
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{\text{osc}} = 10$ oscillations
 - (c) Put a datatip on the plot to identify the maximum displacement and compare with result of item (5)(a)
- (6) Graduate work: calculate the maximum stress at the root of the beam and the safety factor $\text{SF} = Y/\sigma$ and comment on your results for the above two cases:
 - (a) σ_{st} for the static case
 - (b) σ_{dyn} for the dynamic case
- (7) Discuss your results

C.7 Analyze Flex-Beam Pendulum

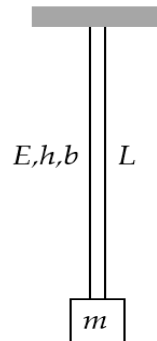


Figure 17: flex-beam pendulum

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 12. The beam is fixed into the ceiling and is hanging downward along the z axis. The mass may oscillate sideways in either x or y directions. The beam cross-section properties are E, h, b . Do the following:

- (1) Calculate the minor and major second moments of area I_1 and I_2
- (2) Assume a horizontal displacement $u(t)$ which produces flexure about the minor I_1 .
- (3) Draw free-body diagram (FBD)
- (4) Write Newton's law of motion (NLM) equation
- (5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (6) Define the natural frequency $\omega_{n,\text{FBP}}$ of the flex-beam pendulum
- (7) Deduce the governing equation
- (8) Compare your results with those of Problem B.1 and write the general solution
- (9) Solve the initial value problem (IVP), *i.e.*, deduce the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$
- (10) Graduate Work: explain how the solution would change if flexure was about I_2

C.8 Calculate Flex-Beam Pendulum Response To Impact

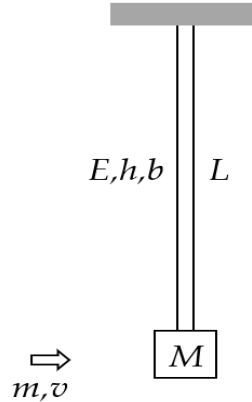


Figure 18: impact excitation of a vertical flex-beam pendulum

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 568 \text{ mm}$ with a mass $M = 42 \text{ kg}$ attached at its end as shown in Figure 18. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The beam is initially at rest. A projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded into it. The azimuthal orientation of the projectile motion is such that the beam flexes about its minor cross-sectional stiffness. This impact imparts an initial velocity to the system that starts to oscillate. Do the following:

- (1) Display input data
- (2) Calculate the minor second moment of area I of the beam
- (3) Calculate the minor cross-sectional stiffness EI
- (4) Calculate the flex-beam spring constant k
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass
- (6) Calculate the rad/s natural frequency $\omega_{n,\text{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it
- (7) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) the largest response
 - (b) the response value calculated at item (7)
- (9) Discuss your results in comparison with results of Problem A.3

C.9 Complex Exponentials And Trigonometric Functions

- (1) Find the solutions of $z_4 = 1$ (aka the fourth roots of unity) and express them as
 - (a) complex numbers
 - (b) complex exponentials
- (2) Write the complex number $e^{i\omega t}$ as a sum of trigonometric functions (hint, Euler's formula)
- (3) Calculate the phase difference ϕ in radians and degrees between the following signals
 - (a) $x_1(t) = \sin(\omega t)$, $x_2(t) = \sin(\omega t + 3\pi/4)$
 - (b) $x_1(t) = \sin(\omega t + \pi/3)$, $x_2(t) = \sin(\omega t + 3\pi/4)$
 - (c) $x_1(t) = \sin(\omega t + 45^\circ)$, $x_2(t) = \sin(\omega t + 115^\circ)$
- (4) What is the phase in radians and degrees of the following complex numbers:
 - (a) $e^{i\pi/2}$
 - (b) $e^{i3\pi/2}$
 - (c) $e^{i\pi/4}$
 - (d) $e^{-i\pi/2}$
 - (e) $e^{-i\pi/4}$
- (5) Draw the locations on the unit circle of the complex numbers in Section (3) above
- (6) Calculate the magnitude and phase in degrees of the complex number $Z = 4+5i$